

Complex Problem Identification

Application domains, identified complex problems, and justification

S.No	Application Domain	Complex Problem Identified	Justification
1	Healthcare Technology	Real-time patient monitoring with wearable devices	Requires integration of IoT sensors, secure data transmission, and AI analysis to detect anomalies instantly. Balancing accuracy, privacy, and scalability makes it complex.
2	Smart Transportation	Autonomous traffic management in urban areas	Involves dynamic coordination of vehicles, pedestrians, and signals using AI. Complexity arises from unpredictable human behavior, high data volume, and safety-critical decisions.
3	Cybersecurity	Detecting zero-day attacks in cloud environments	Needs advanced anomaly detection, machine learning, and adaptive defense mechanisms. The challenge is identifying unknown threats without disrupting legitimate operations.

Note: The problems listed above are complex because they combine technical integration, real-time decision-making, data security, and reliability requirements.